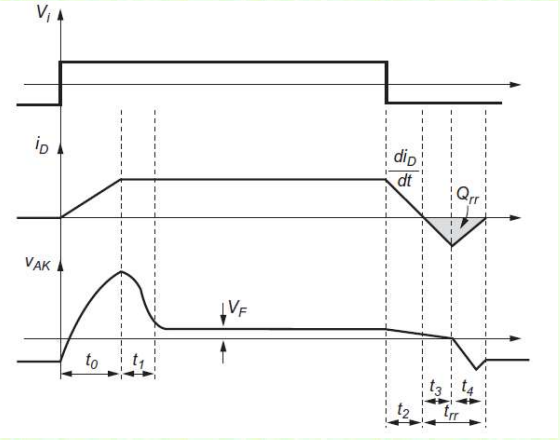
Lecture 4

1

Power Diode
Dynamic Characteristics



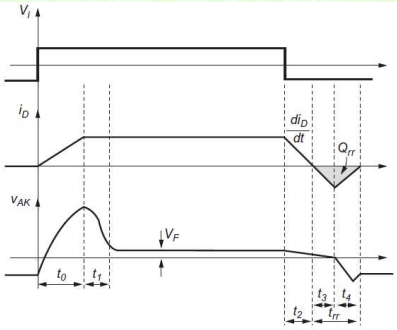
The figure presents a typical dynamic behavior for the voltage and current of a diode during the turn-on and turn-off intervals.

The waveform can change slightly from diode to diode and based on intrinsic properties and also if the circuit contains inductive load.

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Power Diode-Dynamic Characteristics

Time interval t_0: The time interval t_0 begins when the voltage source changes its value from negative to positive, initializing the turn-on process. During t_0, the storage charges in the depletion area are removed. The diode current begins to rise, while the diode voltage suffers a peak due to the fact that there is no enough injection of carriers into the depletion area and also due to the resistance of the N region.

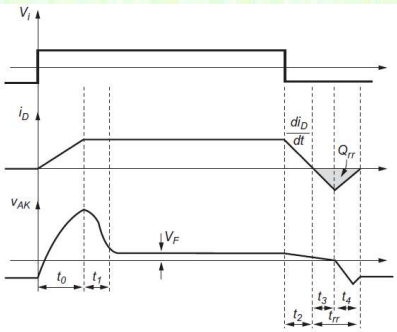


Time interval t_1: At this interval, the injection of carriers is completed, and the diode voltage falls to the forward voltage. In power diodes, the intervals t_0 and t_1, are in the range of ten to hundreds of nanoseconds. At the end of t_1, the turn-on process is completed.

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Power Diode-Dynamic Characteristics

Time interval t_2: The interval t_2 begins when the voltage source changes in value from positive to negative, initializing the turn-off process. The minority carriers (electrons in P region and holes in N region) must be removed before the formation of the potential barrier, during such time the diode keeps conducting. The diode voltage falls slightly due to the fall in the resistance R_{ON}.

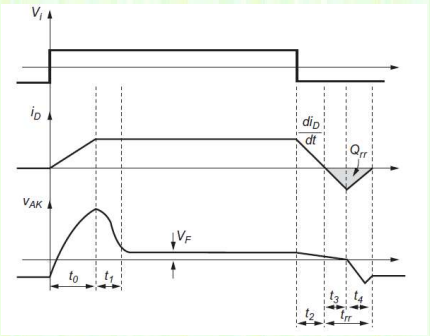


Time interval t_3: The interval t_3 keeps the behavior of the previous interval, but the diode current is achieving negative current. When the current achieves its negative peak, the minority carriers were reversed, and the diode begins to be reversed-biased.

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Power Diode-Dynamic Characteristics

Time interval t_4: The interval t_4 is the time required for the current to achieve null value, recharging the depletion area. The voltage across the diode suffers an overvoltage mainly due to the parasitic inductances found in the circuit



Time interval t_{rr}: The time interval t_{rr} is the sum of the intervals t_3 and t_4 and it is known as reverse recovery time. The area under the t-axis in the current waveform is equivalent to the so-called reverse recovery charge.

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Power Diode-Dynamic Characteristics

The reverse recovery time is an important parameter to decide the switching frequency of applications involving diodes. The reverse recovery time must finish before the diode can be forward-biased again. The lower the t_{rr} is, the faster the diode can be switched. Diodes with low reverse recovery time are known as ***fast diodes*** or sometimes ***ultrafast diodes***.

Another important feature concerning reverse recovery process is how severe the diode current behaves during the reverse recovery time. For this purpose, the terms ***soft- and hard-recovery diodes*** are used. Soft-recovery diodes present slow current fall, while hard-recovery diodes present fast current fall. A slow current fall is more attractive because the voltage produced at the parasitic inductances is lower compared with other case.

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Common Types of Diodes

Small signal diode: This type of diode has a niche of application in low-power rectifiers, limiters, amplifiers, and in waveshaping. These applications are low-voltage, low-current devices. Forward voltage, reverse breakdown voltage, and reverse recovery current and time are parameters that must be known by a designer when applying small signal diode in power electronic applications.



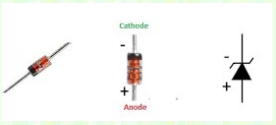


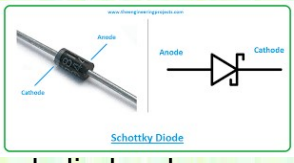
Power rectifier diode: This type of diode has high forward current and high reverse breakdown voltage capabilities. They are suited for power applications like rectifiers, inverters, and power supplies. The thermal resistance is an important factor for a designer, since these diodes usually require heat sink for heat dissipation.

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Common Types of Diodes

Zener diode: This type of diode is mainly used in voltage regulation. It operates on the reverse-biased mode. A current called zener current conducts through it in order to clamp the voltage. The behavior on the breakdown voltage is designed to support the dissipation without damaging the device.



Schottky diodes: The aforementioned diodes have an ohmic behavior in the contact between the metal terminal and the semiconductor. Schottky diodes are formed by allocating a thin film of metal directly to the semiconductor material. As result, the behavior in the referred region has a rectifier effect. The forward voltage drop in Schottky diodes is usually 0.3 V, and they are used in low-voltage applications.

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Power Diode Types

Depending on the recovery characteristics and manufacturing techniques, the power diodes can be classified into the following three categories:

1. Standard or general-purpose diodes
2. Fast-recovery diodes
3. Schottky diodes

General-Purpose Diodes: The general-purpose rectifier diodes have relatively high reverse recovery time, typically 25 μs; and are used in low-speed applications, where recovery time is not critical. These diodes cover current ratings from less than 1 A to several thousands of amperes, with voltage ratings from 50 V to around 5 kV.

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Power Diode Types

Fast-Recovery Diodes: The fast-recovery diodes have low recovery time, normally less than 5 μs. They are used in dc–dc and dc–ac converter circuits, where the speed of recovery is often of critical importance. These diodes cover current ratings of voltage from 50 V to around 3 kV, and from less than 1 A to hundreds of amperes.

Schottky Diodes: In this type of diode, the rectifying action depends on the majority carriers only, and as a result there are no excess minority carriers to recombine. The recovery effect is due solely to the self capacitance of the semiconductor junction. The recovered charge of a Schottky diode is much less than that of an equivalent pn junction diode. Because it is due only to the junction capacitance, it is largely independent of the reverse di/dt.

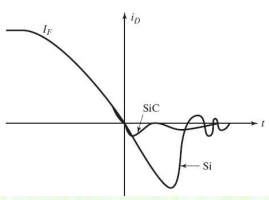
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Silicon Carbide Diodes

Silicon carbide (SiC) is a new material for power electronics. Its physical properties outperform Si and GaAs by far. For example, the Schottky SiC diodes manufactured by Infineon Technologies have ultralow power losses and high reliability. They also have the following features:

- No reverse recovery time;
- Ultrafast switching behavior;
- No temperature influence on the switching behavior.

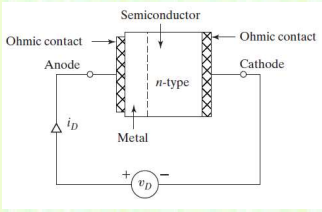
The low reverse recovery characteristic of SiC saves energy in many applications such as power supplies, solar energy conversion, transportations, and other applications such as welding equipment and air conditioners. SiC power devices enable increased efficiency, reduced solution size, higher switching frequency, and produce significant less electromagnetic interference (EMI) in a variety of applications.



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Silicon Carbide Schottky Diodes

Schottky diodes are used primarily in high frequency and fast-switching applications. Many metals can create a Schottky barrier on either silicon or GaAs semiconductors. A Schottky diode is formed by joining a doped semiconductor region, usually n-type, with a metal such as gold, silver, or platinum.



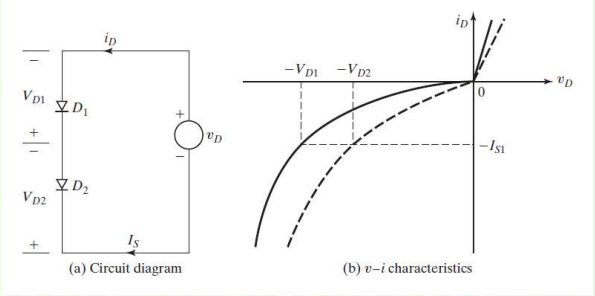


The SiC Schottky diodes have the following features:

- Lowest switching losses due to low reverse recovery charge;
- Fully surge-current stable, high reliability, and ruggedness;
- Lower system costs due to reduced cooling requirements;
- Higher frequency designs and increased power density solutions.
- These devices also have low device capacitance that enhances overall system efficiency, especially at higher switching frequencies.

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Series-Connected Diodes



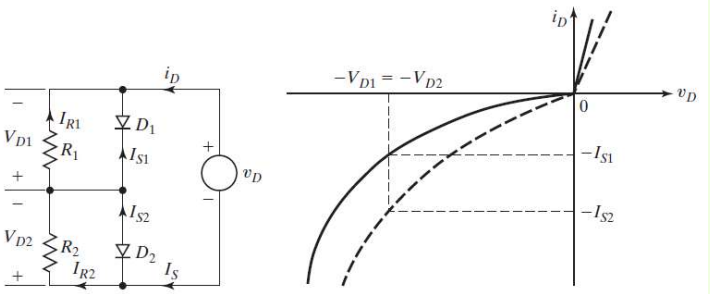
(a) Circuit diagram (b) v–i characteristics

In many high-voltage applications (e.g., high-voltage direct current [HVDC] transmission lines), one commercially available diode cannot meet the required voltage rating, and diodes are connected in series to increase the reverse blocking capabilities.

In practice, the *v-i* characteristics for the same type of diodes differ due to tolerances in their production process. In the reverse blocking condition, each diode has to carry the same leakage current, and as a result the blocking voltages may differ significantly.

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Series-Connected Diodes



(a) Circuit diagram (b) v–i characteristics

A simple solution to this problem, as shown in Figure a, is to force equal voltage sharing by connecting a resistor across each diode. Due to equal voltage sharing, the leakage current of each diode would be different, and this is shown in Figure b. Because the total leakage current must be shared by a diode and its resistor,.

$$-V_{D1} = -V_{D2}$$

$$R = R_1 = R_2$$

$$I_s = I_{S1} + I_{R1} = I_{S2} + I_{R2}$$

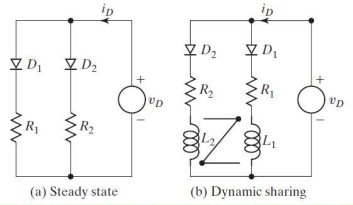
$$I_{S1} + \frac{V_{D1}}{R_1} = I_{S2} + \frac{V_{D1}}{R_2}$$

$$I_{S1} + \frac{V_{D1}}{R} = I_{S2} + \frac{V_{D2}}{R}$$

$$V_{D1} + V_{D2} = V_S$$

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Parallel-Connected Diodes



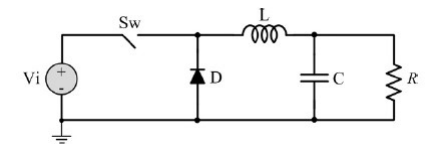
(a) Steady state (b) Dynamic sharing

In high-power applications, diodes are connected in parallel to increase the current carrying capability to meet the desired current requirements. The current sharing of diodes would be in accord with their respective forward voltage drops. Uniform current sharing can be achieved by providing equal inductances (e.g., in the leads) or by connecting current-sharing resistors (which may not be practical due to power losses); this is depicted in Figure. It is possible to minimize this problem by selecting diodes with equal forward voltage drops or diodes of the same type. Because the diodes are connected in parallel, the reverse blocking voltages of each diode would be the same.

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Typical Application of Diodes

Rectifiers and Voltage Multiplier



Freewheeling: A diode can be used as a freewheeling path for keeping an inductor current conducting continuously. Without the diode, the inductor would cause an extremely high dv/dt across it when the transistor changes from ON to OFF state. Such high di/dt would appear across the transistor, probably destructing it. However, the diode allows a path to the inductor current avoiding high di/dt. The voltage produced by the inductor current will be high enough to directly polarize the diode. In this converter, this voltage is just higher than the output voltage. The transistor and the diode are considered ideal.

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